

# Task Expansion and Cross Refinement for Open-World Conditional Modeling

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## Abstract

Open-world conditional modeling (OCM), requires a single model to answer arbitrary conditional queries across heterogeneous datasets, where observed variables and targets vary and arise from a vast open-ended task universe. Because any finite collection of real-world datasets covers only a small fraction of this space, we propose Task Expansion and Cross Refinement (TEXR), a semi-supervised framework that enlarges effective task coverage through structured synthesis and refinement of semantic data contexts. TEXR first generates diverse uninstantiated dataset schemas and weakly instantiates them via structured probabilistic generators guided by large language models. It then performs cross-model refinement by training on disjoint data partitions and revising synthetic values across splits to reduce confirmation bias and improve pseudo-value quality. The refined synthetic datasets are aggregated with real data to train a unified conditional model. Across heterogeneous tabular benchmarks, TEXR consistently improves zero-, few-, and many-shot performance for multiple OCM backbones, demonstrating that structured task expansion and cross refinement enhance open-world conditional modeling.

## 1 INTRODUCTION

Modern foundation models—most prominently large language models (LLMs)—have demonstrated an emerging ability to perform inference across a strikingly broad range of tasks without task-specific retraining, suggesting a pathway toward general-purpose reasoning systems [Brown et al., 2020, Wei et al., 2022]. Inspired by this paradigm, a recent subset of tabular foundation models [Van Breugel and Van Der Schaar, 2024] such as TP-BERTa [Yan et al.,

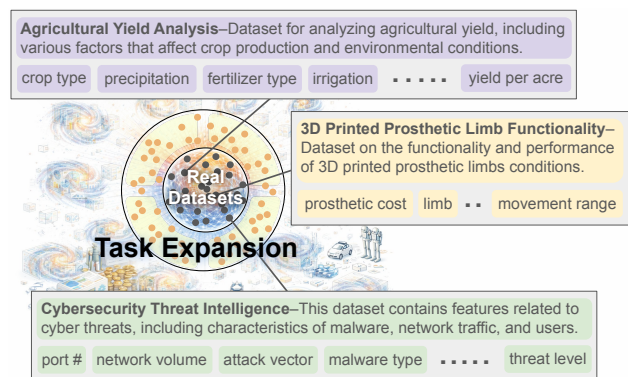


Figure 1: The **task universe** for open-world conditional modeling is vast, and any finite collection of real-world datasets (gray) captures only a small fraction of it. We propose synthetic data generation and refinement techniques to expand task coverage (orange).

2024], CM2 [Ye et al., 2024], GTL [Wen et al., 2024], TabSTAR [Arazi et al., 2026] and ASPIRE [Brahnavar et al., 2025] aim to perform **semantically-grounded open-world conditional modeling** (OCM)<sup>1</sup>. Given *little* (few-shot) or *no* (zero-shot) examples, along with an arbitrary set of observed variables and their semantic descriptions, these models estimate a designated target—even when the feature set or prediction task was never explicitly encountered during training.

This capability is crucial for enabling statistically informed predictions in **data-starved or hypothetical settings**, where labeled data for the exact task may be scarce or unavailable. For example, a single generalist model might be asked to predict: (i) the risk of septic shock from a sparse subset of labs and vitals in a newly integrated hospital system; (ii) regional hospitalization burdens from partial demographic and policy indicators; or (iii) gene-expression responses

<sup>1</sup>Some tabular foundational models do not perform semantically grounded OCM and instead provide a finetuneable or in-context adjusted pretrained model for a new *many-shot* dataset.

from a limited panel of molecular measurements.

However, this *open-world inference problem* is exceptionally difficult because the support of possible tasks spans the combinatorial space of all feature subsets, target choices, and domain-specific data-generating processes. Although there has been a growing effort to aggregate large collections of real-world datasets for pretraining, even expansive collections (spanning thousands of datasets) represent only a **small fraction of this universal task support**. Consequently, models are forced to operate largely in out-of-distribution regimes at test time. Since acquiring real-world data across vast encompassing scenarios is prohibitive, this inherent coverage gap remains a fundamental barrier to open-world conditional modeling (see Fig. 1).

To bridge this gap, we introduce the first principled, semi-supervised framework designed specifically to scale task coverage for OCM without requiring additional real-world data. We begin by recasting OCM as a *context-to-feature* generative problem. Specifically, *we show an equivalence* between training a model to map arbitrary features to arbitrary targets, and training a conditional generator that produces coherent feature values conditioned purely on a semantic data context (table-level and column-level descriptions, variable names, and types). This reframing naturally unlocks a semi-supervised formulation. In addition to fully observed datasets (semantic contexts paired with instance-level values), we can incorporate *context-only specifications*—collections of dataset/variable descriptions and metadata without any associated data rows. These "uninstantiated" contexts define the structure of a task without providing realizations, functioning analogously to unlabeled examples in classical semi-supervised learning.

To construct and leverage the "unlabeled" data (i.e., "uninstantiated" synthetic contexts with no feature values), we introduce **Task Expansion and Cross Refinement (TEXR)**, a semi-supervised framework designed to scale effective task coverage. TEXR first performs *task expansion* by synthesizing diverse, semantically grounded data contexts (schemas with feature and dataset descriptions) and *weakly instantiates* them using structured probabilistic generators guided by large language models. This produces heterogeneous synthetic datasets that represent plausible but previously unseen conditional tasks. It then applies *cross refinement*, where models are trained on disjoint data partitions and used to revise each other's synthetic labels across splits. Finally, the refined synthetic datasets are aggregated with real data to train a unified, open-world conditional model.

**Contributions** Our core contributions are as follows:

- **Semi-Supervised Learning for OCM.** We establish a principled mathematical link between arbitrary open-world prediction and context-conditioned generation, unlocking a semi-supervised pathway for tabular modeling.
- **Synthetic Task Augmentation.** We develop an approach

to scale task coverage by synthesizing diverse, semantically grounded contexts and weakly instantiating them via LLM-guided structured probabilistic generators.

- **Cross-Model Refinement.** We propose a procedure where models trained on disjoint data partitions revise synthetic pseudo-instances to refine their values while reducing error amplification and mitigating confirmation bias.

- **Empirical Experiments.** Across heterogeneous tabular benchmarks, TEXR yields considerable performance gains in zero-, few-, and many-shot regimes for multiple OCM backbones, outperforming existing state-of-the-art synthetic data and pseudo-labeling baselines.

- **A Public Corpus of Synthetic Tasks.** Alongside our code, we release a collection of 10,000+ high-quality, refined synthetic datasets spanning a broad range of scenarios to accelerate future research in open-world tabular inference.<sup>2</sup>

## 2 RELATED WORKS

### 2.1 TABULAR FOUNDATION MODELS

**Transfer Learning for Many-shot Inference.** A central objective in tabular learning is to leverage patterns learned from diverse collections of datasets to improve performance on a new target dataset unseen during pretraining. Unlike conventional per-dataset models (e.g., MLPs or gradient-boosted trees), these approaches aim to transfer statistical structure across tasks.

The TabPFN family [Hollmann et al., 2023, Ye et al., 2026, Grinsztajn et al., 2026], TabICL [QU et al., 2025], and LimiX [Zhang et al., 2025] are strong cross-tabular predictors based on in-context or meta-learning paradigms. These models primarily rely on learned column embeddings and summary-statistic representations derived from feature values, and generally do not incorporate rich natural-language descriptions of datasets or features.

**Semantically-Grounded Tabular Learning.** In parallel, another line of work incorporates textual and metadata descriptions to ground tabular prediction semantically over arbitrary feature-target configurations. This is particularly important in zero- or (very) few-shot OCM settings, where summary statistics alone are insufficient (or unavailable in the zero-shot case) to characterize a new dataset.

CM2 Ye et al. [2024] and TP-BERTa Yan et al. [2024] pretrain shared backbones via cross-table objectives and finetune for downstream tasks. CM2 employs masked table modeling that integrates feature semantics, while TP-BERTa adapts a pretrained language model by fusing textual feature descriptions with discretized numerical values. ASPIRE Brahmavar et al. [2025] extends cross-table masked modeling with semantic conditioning, training transformer architectures that aggregate heterogeneous features while lever-

<sup>2</sup><https://github.com/Shreyas-Bhat/TEXR>

aging dataset- and column-level textual context. TabSTAR Arazi et al. [2026] considers semantically target-aware representations by unfreezing a pretrained text encoder and injecting target tokens as inputs, enabling cross-dataset transfer without dataset-specific output heads. LLM-based approaches such as TabLLM Hegselmann et al. [2023], FeatLLM Han et al. [2024], and GTL Wen et al. [2024] serialize rows into text and cast prediction as conditional language modeling via prompting or supervised fine-tuning. These methods leverage in-context learning and semantic reasoning to generate predictions through next-token likelihood conditioned on feature values and textual context.

*Motivated by low-data OCM, we study how TEXR enhances these semantically grounded cross-tabular models.*

## 2.2 SYNTHETIC DATASET GENERATION FOR CROSS-TABULAR GENERALIZATION

**Synthetic Priors for Cross-Tabular Transfer.** TabPFN-style models [Hollmann et al., 2023, Ye et al., 2026, Grinsztajn et al., 2026] demonstrate that training on large collections of synthetic datasets can induce strong cross-dataset generalization. These models sample tasks from structured priors over functional classes, feature dependencies, noise, and sample regimes, enabling in-context transfer at test time. However, their synthetic data operate over abstract, black-box feature spaces and do not incorporate dataset-level semantics, column descriptions, or natural-language context.

**Semantically-Grounded Synthetic Generation.** Recent work leverages semantic metadata and large language models to generate faithful tabular samples, particularly in low-data settings. Approaches such as StructSynth [Liu et al., 2025], CTSyn [Lin et al., 2025], GReaT [Borisov et al., 2023], TabuLa [Zhao et al., 2025], and CLLM Seedat et al. [2024] condition generation on textual descriptions or schema information *along with a few instances* to produce additional samples. These methods are typically used and evaluated w.r.t. a single target dataset at test time, where synthetic data is assessed for fidelity to that dataset or used to augment and regularize its downstream model, rather than to expand the task distribution underlying cross-dataset pretraining for OCM backbones

*TEXR instead uses synthetic data—starting from a zero-shot, dataset-free formulation—to expand effective task coverage for semantically grounded cross-tabular models, enabling a semi-supervised regime that improves OCM.*

## 2.3 PSEUDO-LABELING FOR TABULAR FOUNDATIONAL MODELS

Pseudo-labeling is a standard semi-supervised strategy for neural networks, originating with [Lee et al., 2013], which treats high-confidence predictions on unlabeled data as train-

ing targets. In tabular learning, these approaches are primarily applied within single-dataset settings to improve pseudo-label reliability rather than cross-dataset open-world generalization [Yoon et al., 2020, Farahani et al., 2022, Kim et al., 2023b]. For example, CAST [Kim et al., 2023a] calibrates pseudo-label confidence using density- and cluster-based signals during self-training on a given table. STUNT [Nam et al., 2023] augments a dataset by appending cluster-derived pseudo-labels to enhance representation learning, rather than expanding the set of datasets into new task scenarios. MediTab [Wang et al., 2024] leverages cross-dataset medical patterns to generate synthetic patients by imputing shared patient characteristics into existing tables. However, MediTab relies on the shared applicability of patient attributes across datasets and does not extend beyond the clinical attribute space into new semantic task scenarios.

*In contrast, TEXR performs instance-level value refinement to instantiate new structured tasks from semantic contexts, expanding the task universe for zero-/few-shot OCM.*

## 3 METHOD

We derive a semi-supervised framework that treats semantic data contexts as the fundamental unit of unsupervised learning for open-world conditional modeling. To address the coverage gap of real-world datasets (Fig. 1), our framework—**Task Expansion and Cross Refinement (TEXR)**—constructs a diverse pool of uninstantiated contexts, generates weak but structured synthetic tabular values, and iteratively refines them through a cross-model co-training procedure. This pipeline expands the model’s coverage over the open-world task distribution while maintaining alignment with real-world data priors.

### 3.1 OCM AS CONTEXT CONDITIONED JOINT MODELING

As aforementioned, recent efforts have resulted in various models capable of performing open-world conditional modeling. Formally, these models aim to learn a parameterized conditional estimate of  $p_\theta(x_t \mid x_o, \mathbf{S})$ ; at inference time, given a semantic data context  $\mathbf{S}$  (e.g., dataset metadata and natural language feature descriptions) and an arbitrary set of observed feature values  $x_o$ <sup>3</sup>, the objective is to estimate a designated target variable  $x_t$ .

To achieve arbitrary conditioning and arbitrary target prediction, open-world conditional models are typically trained based on minimizing the negative log conditional likelihood over a vast task distribution:

$$-\mathbb{E}_{\mathbf{S} \sim \mathcal{T}} \mathbb{E}_{x \sim \mathcal{P}_{\mathbf{S}}} \mathbb{E}_{o, t \sim \mathcal{M}_{\mathbf{S}}} [\log p_\theta(x_t \mid x_o, \mathbf{S})] \quad (1)$$

<sup>3</sup>For notational simplicity, we formulate the zero-shot case here; the few-shot case is analogous but additionally incorporates a support set of examples.

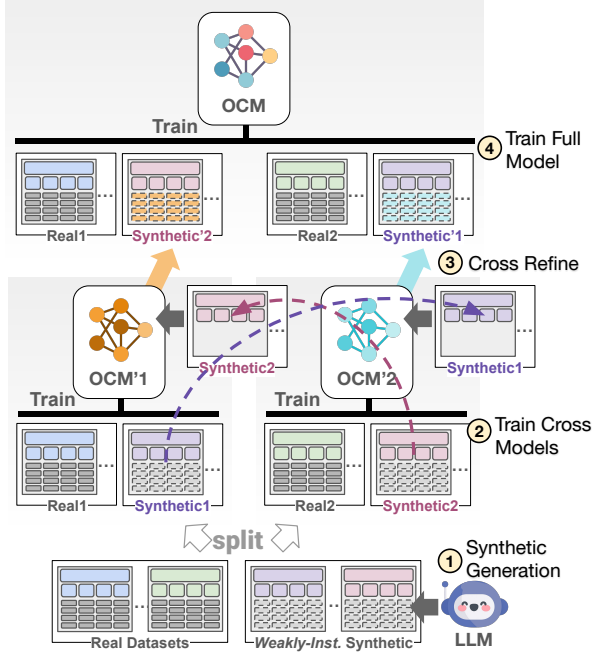


Figure 2: **TEXR** provides an initial collection of weakly-instantiated datasets that are partitioned and then refined with cross model labeling (which refine values on synthetic data not seen during training). Finally, revised synthetic and real datasets are consolidated to train a full open-world conditioning model.

where  $\mathcal{T}$  represents the universal task distribution of semantic contexts,  $\mathcal{M}_S$  is a masking distribution that randomly selects observed features  $o$  and target  $t$ , and  $\mathcal{P}_S$  is the true data distribution from which actual instances  $x$  are sampled.

However, this native formulation is not directly conducive for synthetic data generation and semi-supervised learning since the generation of "unlabeled" examples is *no easier* than the generation of "labeled" examples. That is, the generation of an unlabeled example—a coherent collection of feature descriptions, feature values, and dataset description—is itself a "labeled" example, since we can transfer over any one of the features to act as a target. Below, we mathematically *decouple* the task semantic data descriptions from the observed data values for a formulation that more readily leads to a semi-supervised framework.

**Proposition 3.1** (Informal). Open-world conditional model training on arbitrary inputs and targets is equivalent to training based on context-conditioned joint modeling.

*Proof Sketch.* We note that arbitrary conditioning likelihoods [Strauss and Oliva, 2021]  $\log p_\theta(x_t \mid x_o, \mathbf{S})$  are directly capable of modeling a joint likelihood via the chain rule of probability under an arbitrary feature permutation  $\pi$ :  $\log p_\theta(x \mid \mathbf{S}) = \sum_{i=1}^{d_S} \log p_\theta(x_{\pi_i} \mid x_{\pi_{<i}}, \mathbf{S})$ .

When the masking distribution  $\mathcal{M}_S$  in Eq. (1) uniformly

produces targets and observed feature subsets (of arbitrary cardinality, as is common Ye et al. [2024], Brahmavar et al. [2025]), then the stochastic training objective (1) is mathematically equivalent to minimizing the normalized negative log joint likelihood:

$$- \mathbb{E}_{\mathbf{S} \sim \mathcal{T}} \mathbb{E}_{x \sim \mathcal{P}_S} \left[ \frac{1}{d_S} \log p_\theta(x \mid \mathbf{S}) \right] = \quad (2)$$

$$- \mathbb{E}_{\mathbf{S} \sim \mathcal{T}} \mathbb{E}_{x \sim \mathcal{P}_S} \mathbb{E}_\pi \left[ \frac{1}{d_S} \sum_{i=1}^{d_S} \log p_\theta(x_{\pi_i} \mid x_{\pi_{<i}}, \mathbf{S}) \right]. \quad (3)$$

That is, the standard OCM loss (1) stochastically subsamples the conditionals in the joint factorization (3). Please see Supp. Mat. § 7) for the proof.

This generative reframing (2) clearly delineates the inputs to OC models as the *semantic data contexts*,  $\mathbf{S}$ , and the outputs as the corresponding instance values,  $x$ . Crucially, this equivalence naturally unlocks a semi-supervised learning paradigm. By establishing OCM as a context-conditioned joint generator  $p_\theta(x \mid \mathbf{S})$ , we can formally treat *uninstantiated semantic contexts* (dataset schemas/meta-data with no data rows) as our "unlabeled" instances.

**Semi-Supervised OCM** Our framework (2) implies that we may view OCM as training on a *supervised* collection of instantiated datasets:  $\mathcal{D} = \{(\mathbf{S}^{(k)}, \mathbf{X}^{(k)})\}_{k=1}^N$ , where  $\mathbf{X}^{(k)}$  is a dataset of values for  $n^{(k)}$  instances, corresponding to  $\mathbf{S}^{(k)}$ . Therefore, an *unsupervised* collection would then correspond to a set of  $\mathcal{U} = \{\tilde{\mathbf{S}}^{(k)}\}_{k=1}^M$  of semantic contexts without corresponding dataset values (i.e., uninstantiated semantic contexts). While this opens an avenue for semi-supervised learning, it begs the question of how one may collect an uninstantiated collection  $\mathcal{U}$ . Unlike in classic settings like computer-vision, where unlabeled instances abound, the notion of an uninstantiated dataset is much rarer. Here, we propose to generate a diverse collection of hypothetical data scenarios and respective contexts.

### 3.2 TASK EXPANSION VIA SYNTHETIC CONTEXT

While uninstantiated semantic contexts provide a principled source of unsupervised data for OCM, acquiring them at the scale required to meaningfully expand task coverage is impractical from real-world sources alone. We therefore introduce *Task Expansion*: a systematic synthesis of diverse dataset contexts using large language models (LLMs), leveraging their broad semantic priors. We construct the unsupervised context pool through a hierarchical two-stage procedure (see Supp. Mat. § 8):

1. *Topic Universe Construction.* To promote broad domain coverage, we prompt an LLM to first enumerate high-level domains (e.g., *Technology & Digital, Health & Sciences, Finance & Business*), decompose them into

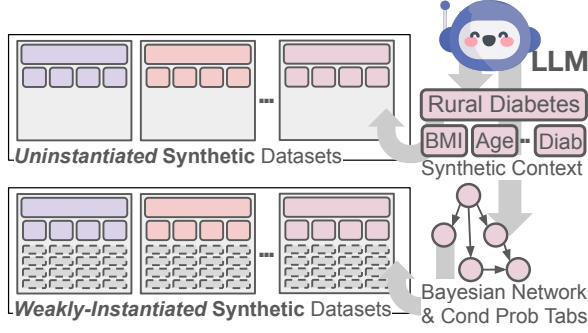


Figure 3: Our initial collection of **synthetic datasets** stems from an LLM guided process that first generates hypothetical data contexts followed by a weak-labeling scheme that generates corresponding Bayesian networks and conditional probability tables to produce instance values.

sub-fields, and finally generate concrete predictive scenarios (e.g., *hospital readmission risk*, *customer churn*, *loan default prediction*). This yields a large topic set  $\mathcal{T} = \{\tau_k\}_{k=1}^K$  spanning thousands of hypothetical tasks.

2. *Schema Generation*. For each topic  $\tau \in \mathcal{T}$ , we prompt the LLM to produce a structured (e.g., JSON) semantic context, consisting of: (i) a natural-language *dataset description* summarizing the overarching task and data, and (ii) a feature list in which each variable is paired with a data type and a detailed semantic description.

This procedure yields a large collection of uninstantiated semantic contexts,  $\mathcal{U} = \{\mathbf{S}^{(k)}\}_{k=1}^M$ , each containing rich dataset-/column-level descriptions but no associated rows. Thus,  $\mathcal{U}$  constitutes a purely unsupervised pool of candidate tasks derived from LLM priors. In the following section, we develop a pseudo-labeling, cross-model refinement procedure that leverages  $\mathcal{U}$  for semi-supervised OCM training.

As an illustrative example, Fig. 3 sketches this generative process for a “rural diabetes” scenario: task expansion yields a context with a dataset description and features such as such as BMI, age, and diagnosis, each paired with a type and semantic description. See Supp. Mat. §9 for a walkthrough of this example through weak instantiation and cross refinement.

### 3.3 PSEUDO-VALUE GENERATION AND CROSS REFINEMENT

For OCM training that leverages the unsupervised pool of synthetic contexts,  $\mathcal{U}$ , we introduce a semi-supervised procedure that sequentially couples weak data instantiation with a partition-induced cross-refinement strategy. This phase fills (instantiates) our empty schemas with structured initial values and systematically revises them.

#### 3.3.1 Weak Instantiation and Filtration

We begin by assigning preliminary values to the uninstantiated semantic contexts  $\mathcal{U}$ . Since generating realistic open-world values is the ultimate goal of the OCM itself, we instead synthesize weak but structured statistical signals to bootstrap training. To ensure these synthetic signals provide complementary information, they are generated independently of any observed instances in the real datasets  $\mathcal{D}$ .

For each semantic context, we formulate a plausible probabilistic data-generating process by prompting an LLM to: (i) construct a Bayesian network, a directed acyclic graph (DAG), encoding conditional dependencies between the features, and (ii) a set of corresponding per-node conditional distributions in the form of conditional probability tables. The LLM is prompted once to generate the DAG via a topologically ordered list of features and corresponding edge list, which is post-processed to enforce acyclicity. The conditional probability tables perform a look-up based on parents’ values (for discrete parents) or quartiles (for continuous parents). For discrete nodes, the conditional probability tables directly specifies a pmf over domains. For continuous nodes, the conditional probability tables specifies a pdf as Gaussian mean and variance. Sampling from these graphical models yields our initial pool of weakly instantiated synthetic datasets,  $\mathcal{W} = \{(\tilde{\mathbf{S}}^{(k)}, \tilde{\mathbf{X}}^{(k)})\}_{k=1}^M$ . (Please find further details in Supp. Mat. §9)

Since these graphical structures and conditional probabilities are purely hypothetical, we propose to filter the resulting values for both plausibility and informativeness. We train an ensemble of two preliminary OC models on disjoint partitions of the real datasets  $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$ ,  $\mathcal{D}_1 \cap \mathcal{D}_2 = \emptyset$ . For each synthetic value  $\tilde{x}_j^{(k,i)}$  (the  $j^{\text{th}}$  feature of the  $i^{\text{th}}$  instance in the  $k^{\text{th}}$  dataset), we compute the predictive mean  $\bar{x}_j^{(k,i)}$  and standard deviation  $\sigma_j^{(k,i)}$  from the two real-data-only OC models, conditioned on the remaining features  $\tilde{x}_{-j}^{(k,i)}$ . We then calculate the absolute  $z$ -score:

$$z_j^{(k,i)} = \left| \tilde{x}_j^{(k,i)} - \bar{x}_j^{(k,i)} \right| / \sigma_j^{(k,i)}. \quad (4)$$

This metric elegantly captures two desiderata: a small numerator indicates the value is plausible (it aligns with the real-data models’ expectations), while a large denominator indicates the value is informative (it lies in a region of high predictive uncertainty). Synthetic datasets exhibiting low average feature-level  $z$ -scores are retained.

#### 3.3.2 Cross-Model Pseudo-Value Refinement

Although filtered, the weakly generated synthetic values are not data-grounded and could be refined to produce better training signals. To strengthen pseudo-instance values while mitigating confirmation bias, we employ a novel cross-model refinement procedure.

We partition the available data into two disjoint splits and train two independent OCMs:  $p'_1$  is trained on  $\mathcal{D}_1 \cup \mathcal{W}_1$ , and  $p'_2$  is trained on  $\mathcal{D}_2 \cup \mathcal{W}_2$ . We then perform cross-dataset refinement:  $p'_1$  recalculates and refines the synthetic values in  $\mathcal{W}_2$ , while  $p'_2$  revises those in  $\mathcal{W}_1$ , yielding the finalized, high-quality synthetic collections  $\mathcal{W}'_1$  and  $\mathcal{W}'_2$  (see *Step 3*, Fig. 2). That is, refined  $\mathcal{W}'_1$  contains datasets  $\tilde{\mathbf{X}}_1'^{(k)} \sim p'_2(\cdot \mid \tilde{\mathbf{S}}_1^{(k)})$ , and vice-versa,  $\mathcal{W}'_2$  has datasets  $\tilde{\mathbf{X}}_2'^{(k)} \sim p'_1(\cdot \mid \tilde{\mathbf{S}}_2^{(k)})$  using the autoregressive, context conditioned OCM generation (3).

Since partitioning induces distinct exposure to dataset patterns and semantic scenarios, the two models,  $p'_1$  and  $p'_2$ , develop diverse predictive biases. By ensuring each model only revises the pseudo-values of synthetic datasets it did *not* observe during training, this cross-revision mechanism leverages that diversity to stabilize the estimates and prevent models from reinforcing their own local generative errors.

### 3.4 CONSOLIDATED MODEL TRAINING

In the final stage of TEXR, the refined synthetic collections,  $\mathcal{W}'_1$  and  $\mathcal{W}'_2$ , are joined with the original real-world datasets,  $\mathcal{D}$ , to train a unified open-world conditional model  $p_{\text{full}}$  on the consolidated corpus  $\mathcal{D} \cup \mathcal{W}'_1 \cup \mathcal{W}'_2$  (*Step 4*, Fig. 2). By optimizing the arbitrary conditional masking objective (Eq. 1) over this expanded task universe, the model mitigates the open-world coverage gap, yielding stronger generalization on zero- and few-shot inference over previously unseen semantic contexts (see below).

## 4 EXPERIMENTS

We evaluate OCM cross-tabular models under three deployment regimes: **(I) zero-shot**, **(II) few-shot**, and **(III) many-shot inference**. Evaluation follows Ye et al. [2024] and includes 20 heterogeneous tabular datasets spanning binary classification, multiclass classification, and regression, drawn from UCI, OpenML, and Kaggle.

For synthetic pretraining, we sample 10,000 datasets from our generative pipeline, with all generators constrained to the same generative computational budget. The real-dataset collection,  $\mathcal{D}$ , is the 1,400 OpenTabs datasets [Ye et al., 2024], each paired with natural language dataset- and feature-level descriptions.

**OC Model Backbones.** We assess how data expansion affects various open-world conditional modeling cross-tabular backbones: ASPIRE [Brahmavar et al., 2025], CM2 [Ye et al., 2024], TABSTAR [Arazi et al., 2026], and TP-BERTA [Yan et al., 2024].

**LLM Backbones.** Please see Supp. Mat. § 9 for full details of all LLMs used. TEXR relies exclusively on open-weight LLMs. For competing baselines, we follow the origi-

nally recommended backbones, additionally evaluating updated variants when available and also used within TEXR.

### 4.1 EXPERIMENTAL BASELINES

As no prior work evaluates our full problem formulation, we construct strong baselines by composing state-of-the-art components across the pipeline. To our knowledge, these combinations have not been systematically studied to improve OCM. We factor baselines along two axes: *synthetic task generation* and *pseudo-value refinement*.

**Training Protocol** OC models incorporate synthetic data in a pre-training step before continuing their OCM training on OpenTabs real datasets (see Supp. Mat §8). For each CM backbone, each baseline creates a pipeline by selecting (i) a synthetic task generator, (ii) a pseudo-value refinement strategy. First, the generator produces a pool of weakly instantiated synthetic datasets (analogously to § 3.2). Next, the refinement strategy updates these values using the OCM backbone trained on the weakly instantiated synthetic data (analogously to § 3.3.2). Finally, the full OC model is trained on the union of refined synthetic and real data and evaluated on held-out test datasets.

**Synthetic Task Generators.** We compare the following four generators under a fixed generation budget (further details in Supp. Mat. § 9):

**(i) Our Task Expansion and Weak-Instantiation:** We generate 500 dataset topics (§ 3.2) and sample 20 semantic data contexts per topic (including textual dataset descriptions, textual features descriptions, feature-types). For each feature set, we prompt an LLM once to propose a global edge list in topological order. Edges that introduce cycles are discarded to ensure a valid DAG. We then generate 1,000 rows via topological sampling over the resulting DAG using the generated conditional probability tables (§ 3.3.1), producing weak instantiations of the semantic contexts.

**(ii) StructSynth Weak-Instantiation:** As an alternative weak-instantiation baseline, we use STRUCTSYNTH [Liu et al., 2025]. Because STRUCTSYNTH does not perform task expansion, we apply it to initialize values for semantic data contexts generated in § 3.2. STRUCTSYNTH originally synthesizes instances via LLM-guided prompting conditioned on a few-shot support set and dataset context. In our setting, we remove the support set and prompt the LLM using only our generated dataset contexts. While both STRUCTSYNTH and our weak-instantiation (§ 3.3.1) construct intermediate DAGs, their procedures differ. STRUCTSYNTH incrementally builds the DAG via iterative breadth-first LLM queries, conditioning on the current graph state, and subsequently autoregressively generates each feature conditioned on its parents through LLM generation. In contrast, TEXR generates the full DAG in a single query and samples values from the generated conditional probability tables (§ 3.3.1).

As in [Liu et al., 2025], all generated values are validated and coerced to the inferred schema for STRUCTSYNTH (e.g., categorical values mapped to valid categories, continuous values clipped to valid ranges).

**(iii) CLLM Weak-Instantiation:** We consider the Curated LLM (CLLM) pipeline of Seedat et al. [2024] as an additional weak-instantiation baseline. In its original form, CLLM generates synthetic rows from an in-context few-shot support set followed by a curation stage. Since CLLM is not designed for zero-shot task expansion, we adapt it by removing the support set and instead providing the same hypothetical dataset contexts generated in § 3.2. Following Seedat et al. [2024], we retain the curation mechanism: an XGBoost model scores each synthetic instance using prediction confidence and aleatoric uncertainty, and only the curated subset is retained for downstream training.

**(iv) TabPFN-based Data Expansion:** TabPFN-style synthetic generation has been shown to improve cross-tabular generalization [Hollmann et al., 2023, Ye et al., 2026, Grinsztajn et al., 2026]. Within the TEXR formulation, it naturally provides task expansion by generating entirely synthetic datasets without reliance on real data. However, TabPFN operates over abstract feature spaces and does not produce semantic data contexts, instead generating fully instantiated tables with unnamed, uninterpretable columns. To adapt TabPFN for our OCM setting, we pair its synthetic datasets with TABMETA [Anonymous, 2024] to construct corresponding plausible table- and column-level descriptions. Following the TabPFN generation procedure [Hollmann et al., 2023], datasets are sampled from structural causal model (SCM) priors that induce random DAGs with nonlinear structural equations, where values are produced via forward propagation through randomly parameterized neural components. Because these priors yield structurally valid but semantically ungrounded features, we generate corresponding semantic data contexts post hoc via TABMETA-style prompting. Multiple LLM judges score candidates for relevance, factuality, and coherence, and we select the highest-scoring annotations via majority voting. This produces synthetic datasets with semantic data contexts while leaving the underlying synthetic feature values unchanged.

**Pseudo-value Refinement.** We similarly compare three pseudo-value refinement techniques to update weakly instantiated synthetic values. A final full model (of corresponding OCM backbones) is trained using all real datasets and synthetic value refined data.

**(a) Our Cross-Model Refinement:** We train two OCM models on disjoint partitions of real and weakly instantiated synthetic data to iteratively refine synthetic values (see § 3.3.2, Step 3 of Fig. 2). Refined values are generated autoregressively, with each feature conditioned on previously generated features within the same instance.

**b) Direct Refinement (DR):** As a baseline, we consider a single-model pseudo-value refinement strategy. Instead of cross-refinement, we train one OCM model  $p'$  on the full

set of real and weakly instantiated synthetic data,  $\mathcal{D} \cup \mathcal{M}$ . After training,  $p'$  is used to re-generate (refine) the synthetic values, producing an updated collection  $\mathcal{M}'$  with instances sampled autoregressively as  $\tilde{\mathbf{X}}'^{(k)} \sim p'(\cdot | \tilde{\mathbf{S}}^{(k)})$ .

**c) Cluster-Aware Self-Training (CAST):** We additionally compare against CAST Kim et al. [2023a], a recent tabular pseudo-labeling method that recalibrates prediction confidence using density regularization. As with Direct Refinement, we use a single refinement model (no cross-modeling), but employ CAST’s density-aware confidence scores to guide value refinement, using these calibrated confidences when sampling refined values to better propagate instance-level uncertainty to the final model.

## 4.2 OCM WITH DATA EXPANSION

**Few-shot OCM.** We evaluate data expansion strategies under few-shot inference ( $\leq 5$  labeled examples), following each backbone’s protocol. Fig. 4 reports five-shot F1 across backbones. All expansion methods outperform real-data-only training (NONE), which only trains on the OpenTabs collection; however, TEXR achieves the largest gains, improving performance by an average of  $\Delta=9.8$  percentage points over real-data-only (NONE) across models. As shown in Fig. 5, TEXR consistently provides the strongest and most stable improvements as the number of shots varies. Similar trends hold for regression (Supp. Mat. Fig. 10), where TEXR yields the largest average RMSE reduction.

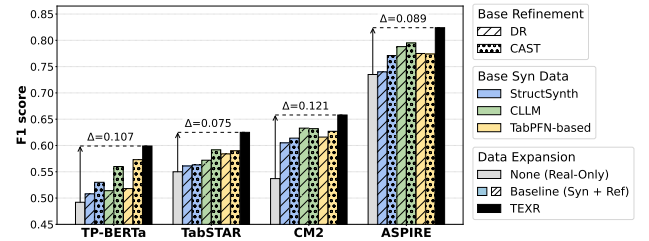


Figure 4: **Five-shot F1 comparison** across OCM backbones and data expansion strategies: synthetic generator (color), refinement (hatch), real-data-only (gray), and TEXR (black).

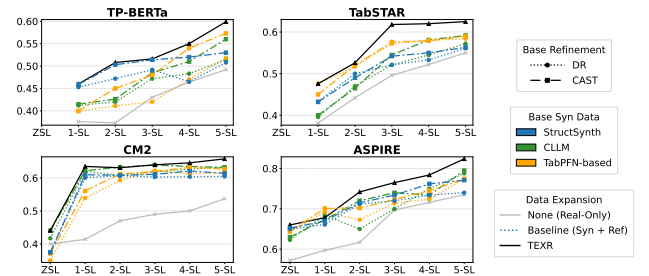


Figure 5: **Zero-shot (ZSL) to Five-shot (5SL) F1 comparison** across data expansion strategies and OCM backbones. (TP-BERTa and TabSTAR are unable to perform ZSL.)

**Many-shot OCM.** Although TEXR is motivated by data-limited regimes, it also improves performance in many-shot conditioning, where backbones observe thousands of labeled inference-time examples under their standard adaptation protocols. Fig. 6 shows consistent gains over real-data-only training, with TEXR improving F1 by an average of  $\Delta=5.38$  percentage points across backbones. This is notable given that, in the many-shot setting, models rely less on cross-tabular transfer due to abundant task-specific supervision. Similar patterns hold for many-shot regression (Supp. Mat. Fig. 11), where TEXR yields the largest average RMSE reduction.

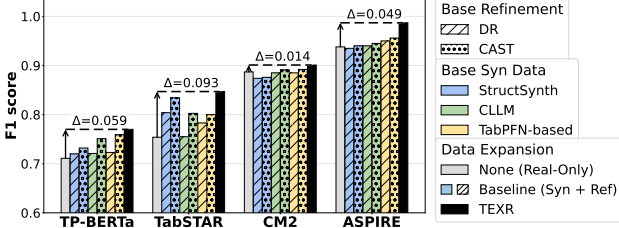


Figure 6: **Many-shot F1 comparison** across data expansion strategies over various OCM backbones.

### 4.3 ABLATIONS

We further analyze TEXR via targeted ablations of its task expansion and pseudo-value refinement components, focusing on the ASPIRE and TABSTAR backbones (Fig. 7 and Fig. 8). First, fixing cross-refinement (§3.3.2), we vary the expansion generator among STRUCTSYNTH, CLLM, and TABPFN-based synthesis. Second, fixing task expansion (§3), we compare refinement strategies: no refinement, direct refinement, and CAST. The full pipeline (TE+XR) achieves the largest gain over real-data-only (NONE). Our task expansion (TE) outperforms alternative generators (\*+XR), highlighting the strength of our synthesis and weak-instantiation design. Refinement—especially cross-refinement—further improves over weak instantiation (TE+NoR), confirming its added OCM value. **z-score filtering:** removing the filter in Eq. (4) reduces performance from 0.824 to 0.815, indicating it helps select informative synthetic tasks.

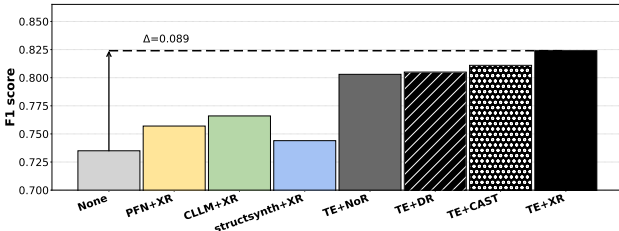


Figure 7: **Data expansion ablation** on ASPIRE.

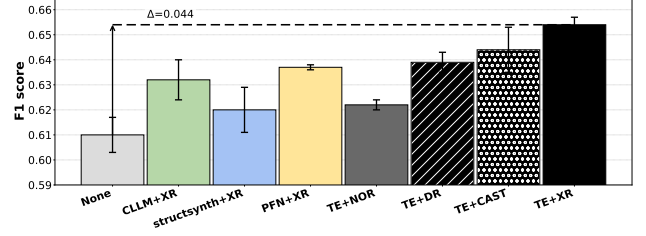


Figure 8: **Data expansion ablation** on TABSTAR.

### 4.4 DIRECT SYNTHETIC DATA QUALITY

We assess both the intrinsic properties and predictive utility of the generated datasets. We evaluate *schema quality*: whether the generated dataset description, feature set, and feature types specify a coherent and domain-appropriate predictive task. To do so, we use a fixed rubric-based LLM judge (Gemini 3 Flash) following the LLM-as-a-judge evaluation protocols [Anonymous, 2024, Chen et al., 2024]. The judge receives the dataset description, feature metadata, valid feature domains, summary statistics, and pairwise feature relationships, while method identities are masked. It assigns scores from 0 to 10 along three axes (Table 1): (i) *schema quality*, measuring semantic coherence, specificity, and compatibility between the task description and feature schema; (ii) *statistical quality*, measuring plausibility, diversity, and non-degeneracy of the generated values and their dependencies; and (iii) *anticipated usefulness*, measuring whether the dataset provides informative signal for conditional prediction. We note that TEXR scores highest on all three axes. The largest margins are on statistical quality (8.47 vs. 7.02–7.51) and anticipated usefulness (8.33 vs. 6.36–7.80), indicating that the Bayesian-network instantiation produces value dependencies that are both more plausible and more informative than direct LLM row generation.

For predictive utility, follow protocol from [Esteban et al., 2017]. A classifier is trained using a fixed subset of real training examples augmented with an equal number of synthetic examples (a ratio of 0.5), and is evaluated exclusively on held-out real test data (Table 2). This setting measures whether the synthetic samples provide complementary predictive information beyond the available real training data. TEXR augmentation yields the largest gain (+0.049 F1 over the real-only baseline), roughly double that of CLLM (+0.021) and StructSynth (+0.029), confirming that its synthetic values carry complementary predictive signal beyond schema-level plausibility—consistent with the usefulness rankings in Table 1.

**Compute and generation cost.** TEXR was designed with practical cost in mind by separating generation into parallelizable, one-time preprocessing stages and reusable downstream training data. In particular, schema generation is an offline cost: dataset descriptions and feature lists are gener-

Table 1: Direct evaluation of synthetic dataset quality. Scores are on a 0–10 scale; higher is better.

| Method       | Schema      | Statistical | Usefulness  | Average     |
|--------------|-------------|-------------|-------------|-------------|
| TEXR         | <b>9.33</b> | <b>8.47</b> | <b>8.33</b> | <b>8.71</b> |
| CLLM         | 8.27        | 7.02        | 7.24        | 7.51        |
| TabPFN-based | 8.45        | 7.21        | 6.36        | 7.34        |
| StructSynth  | 8.47        | 7.51        | 7.80        | 7.59        |

Table 2: Within-dataset predictive utility on held-out real data.

| Training Data            | Synthetic Ratio | F1           | $\Delta$ F1   |
|--------------------------|-----------------|--------------|---------------|
| Real-only half-data      | 0.0             | 0.710        | —             |
| CLLM augmentation        | 0.5             | 0.731        | +0.021        |
| StructSynth augmentation | 0.5             | 0.739        | +0.029        |
| TEXR augmentation        | 0.5             | <b>0.759</b> | <b>+0.049</b> |

ated independently across topics in parallel, with multiple candidate schemas produced per prompt through bulk generation. In our implementation on a single NVIDIA L40S GPU with parallel CPU workers, generation takes roughly 1.33 seconds per dataset—0.79 seconds for schema generation and 0.54 seconds for weak instantiation—so the full released corpus of 10,000 datasets requires only about 3.7 GPU-hours. By comparison, direct LLM-based row generation is far more expensive: CLLM requires approximately 170 seconds per dataset ( $\sim 128\times$  slower). The TabPFN-based generator avoids LLM row generation and is efficient, but at roughly 2 seconds per dataset is still slower than our procedure. Because this cost is incurred once and the refined corpus is released, it is further amortized across OCM backbones and downstream experiments.

## 5 DISCUSSION

**A new semantically-grounded tabular prior.** TabPFN’s synthetic-data paradigm [Hollmann et al., 2023, Grinsztajn et al., 2026] is grounded in a structured generative prior over latent causal graphs, feature marginals, target mechanisms, and noise processes; its inductive bias is explicitly statistical and row-level, shaped by synthetic task distributions that approximate classical machine-learning datasets, yielding strong cross-tabular performance. In contrast, we introduce a *complementary* open-world semantically-grounded prior for generating weakly-instantiated synthetic datasets that encodes broad knowledge of domains, entities, and attribute relationships. As a proper cross tabular instance prior, our LLM-based prior remains largely independent of structured feature–value matrices. This follows from TEXR’s reliance on generalist LLMs, which are pretrained on massive natural-language corpora (e.g., web text, books, code)

rather than curated tabular row datasets [Brown et al., 2020, Touvron et al., 2023]. The tight coupling between semantics and plausible weak instance values makes our dataset prior especially adept at improving open-world conditional modeling, as evidenced by the results above.

**Release of Synthetic Datasets.** In addition to our code, we publicly release<sup>4</sup> a collection of 10,000+ *refined* synthetic datasets (§ 3.3.2) spanning a broad range of topics (Fig. 9, left). As supported by our results, these datasets improve multiple OCM backbones, suggesting that they encode useful structural patterns coupled with semantic data contexts that complement large real-world collections such as OpenTabs [Ye et al., 2024]. A t-SNE visualization (Fig. 9, right) indicates that task expansion increases coverage of regions underrepresented in OpenTabs. We hope this released collection—and the accompanying pipeline—serves as a practical resource for cross-tabular research.

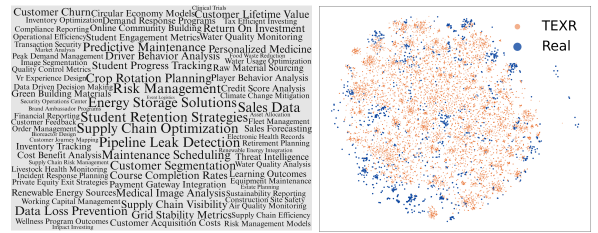


Figure 9: Left, representative **word cloud** of our synthetic datasets. Right, t-SNE visualization of **space of synthetic vs. real data contexts**.

## 6 CONCLUSION

In conclusion, TEXR provides a semi-supervised framework for open-world conditional modeling that expands effective task coverage through structured task expansion and cross-model refinement. By treating semantic data contexts as generative inputs, “unlabeled” contexts can be transformed into weakly instantiated tabular tasks and refined into useful training signal. This yields a novel, complementary semantic prior to existing statistically grounded synthetic paradigms. Across heterogeneous backbones, TEXR improves zero-/few- and many-shot performance, and our release of 10,000+ refined synthetic datasets offers a practical resource for future work on cross-dataset tabular learning. Potentially fruitful future directions include the integration of multiple data expansion strategies jointly, an active-learning search over task expansion, and extensions into multiple modalities.

<sup>4</sup><https://github.com/Shreyas-Bhat/TEXR>

## Acknowledgements

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## Supplementary Material

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### 7 EQUIVALENCE OF OPEN-WORLD CONDITIONAL MODELING AND JOINT GENERATION

Open-world conditional models are trained by minimizing the expected negative log conditional likelihood over a vast task distribution:

$$\mathcal{L}_{\text{OCM}} = -\mathbb{E}_{S \sim \mathcal{T}} \mathbb{E}_{x \sim \mathcal{P}_S} \mathbb{E}_{o, t \sim \mathcal{M}_S} [\log p_\theta(x_t | x_o, S)] \quad (5)$$

where  $\mathcal{T}$  represents the universal task distribution of semantic contexts,  $\mathcal{P}_S$  is the true data distribution for actual instances  $x$ , and  $\mathcal{M}_S$  is a masking distribution that selects observed features  $o$  and target  $t$ . To justify our semi-supervised context-to-feature generative formulation, we establish the conditions under which this objective is equivalent to joint log-likelihood minimization.

**Proposition 7.1.** *Let the masking distribution  $\mathcal{M}_S$  operate by uniformly sampling a feature permutation  $\pi$  from the set of all possible permutations  $\Pi$  of the  $d_S$  features, and uniformly sampling a target index  $i \sim U(\{1, \dots, d_S\})$ , such that the target is  $t = \pi_i$  and the observed set is  $o = \pi_{<i}$ . Under this condition, minimizing the expected OCM objective (Eq. 1) is equivalent to minimizing the normalized negative log joint likelihood of the data conditioned on the semantic context:*

$$\mathcal{L}_{\text{Joint}} = -\mathbb{E}_{S \sim \mathcal{T}} \mathbb{E}_{x \sim \mathcal{P}_S} \left[ \frac{1}{d_S} \log p_\theta(x | S) \right] \quad (6)$$

**Proof.** By the definition of the uniform masking distribution  $\mathcal{M}_S$ , the expectation over the mask can be expanded as the average over all permutations  $\pi \in \Pi$  and all valid target indices  $i$ :

$$\mathbb{E}_{o, t \sim \mathcal{M}_S} [\log p_\theta(x_t | x_o, S)] = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \frac{1}{d_S} \sum_{i=1}^{d_S} \log p_\theta(x_{\pi_i} | x_{\pi_{<i}}, S)$$

By the chain rule of probability, for *any* given permutation  $\pi$ , the sum of the log conditionals factorizes the joint log-likelihood:

$$\sum_{i=1}^{d_S} \log p_\theta(x_{\pi_i} | x_{\pi_{<i}}, S) = \log p_\theta(x | S) \quad (7)$$

Substituting Eq. 7 into the expanded expectation collapses the permutation average:

$$\frac{1}{|\Pi|} \sum_{\pi \in \Pi} \frac{1}{d_S} \log p_\theta(x | S) = \frac{1}{d_S} \log p_\theta(x | S)$$

Substituting this result back into Eq. 1 yields the context-conditioned joint formulation (Eq. 6), establishing equivalence. ■

**Approximate Stochastic Equivalence:** While Proposition 7.1 establishes equivalence under full expectation, practical OCM architectures implement a stochastic approximation. During mini-batch optimization, sampling a single mask  $o, t \sim \mathcal{M}_S$  per instance acts as an unbiased Monte Carlo estimator of the scaled joint log-likelihood gradient. This justifies treating uninstantiated semantic schemas as “unlabeled” instances for semi-supervised generation.

Table 3: Table of the downstream datasets in our experiments, along with different information

| Dataset Name     | R/C | Samples | Numerical | Categorical | Label Classes | Source                                                                                                                                                        |
|------------------|-----|---------|-----------|-------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Breast           | C   | 699     | 9         | 0           | 2             | <a href="https://archive.ics.uci.edu/dataset/15/breast+cancer+wisconsin+original">https://archive.ics.uci.edu/dataset/15/breast+cancer+wisconsin+original</a> |
| Bone             | C   | 1479    | 2         | 7           | 3             | <a href="https://archive.ics.uci.edu/dataset/3/connectionist+bench+choice">https://archive.ics.uci.edu/dataset/3/connectionist+bench+choice</a>               |
| Diabetes         | C   | 768     | 8         | 0           | 2             | <a href="https://openml.org/d/37">https://openml.org/d/37</a>                                                                                                 |
| Vehicle          | C   | 846     | 18        | 0           | 4             | <a href="https://archive.ics.uci.edu/dataset/149/statlog+vehicle+silhouettes">https://archive.ics.uci.edu/dataset/149/statlog+vehicle+silhouettes</a>         |
| Satimage         | C   | 6430    | 36        | 0           | 6             | <a href="https://archive.ics.uci.edu/dataset/146/statlog+landsat+satellite">https://archive.ics.uci.edu/dataset/146/statlog+landsat+satellite</a>             |
| Sick             | C   | 3772    | 7         | 22          | 2             | <a href="http://archive.ics.uci.edu/dataset/102/thyroid+disease">http://archive.ics.uci.edu/dataset/102/thyroid+disease</a>                                   |
| Analcatdata      | C   | 797     | 0         | 4           | 6             | <a href="https://pages.stern.nyu.edu/~jsimonof/AnalCatData/Data/">https://pages.stern.nyu.edu/~jsimonof/AnalCatData/Data/</a>                                 |
| Pcl              | C   | 1109    | 21        | 0           | 2             | <a href="https://openml.org/d/1068">https://openml.org/d/1068</a>                                                                                             |
| Adult            | C   | 48842   | 6         | 8           | 2             | <a href="https://archive.ics.uci.edu/dataset/2/adult">https://archive.ics.uci.edu/dataset/2/adult</a>                                                         |
| PhishingWebsites | C   | 11055   | 0         | 30          | 2             | <a href="https://archive.ics.uci.edu/dataset/327/phishing+websites">https://archive.ics.uci.edu/dataset/327/phishing+websites</a>                             |
| Cylinder-bands   | C   | 540     | 18        | 21          | 2             | <a href="https://archive.ics.uci.edu/dataset/32/cylinder+bands">https://archive.ics.uci.edu/dataset/32/cylinder+bands</a>                                     |
| MiceProtein      | C   | 1080    | 77        | 4           | 8             | <a href="https://archive.ics.uci.edu/dataset/342/mice+protein+expression">https://archive.ics.uci.edu/dataset/342/mice+protein+expression</a>                 |
| Car              | C   | 1728    | 0         | 6           | 4             | <a href="https://archive.ics.uci.edu/dataset/19/car+evaluation">https://archive.ics.uci.edu/dataset/19/car+evaluation</a>                                     |
| Segment          | C   | 2310    | 19        | 0           | 7             | <a href="http://archive.ics.uci.edu/dataset/50/image+segmentation">http://archive.ics.uci.edu/dataset/50/image+segmentation</a>                               |
| Porto-seguro     | R   | 2000    | 26        | 31          | 2             | <a href="https://openml.org/d/44787">https://openml.org/d/44787</a>                                                                                           |
| Amazon           | C   | 2000    | 0         | 9           | 2             | <a href="https://openml.org/d/44712">https://openml.org/d/44712</a>                                                                                           |
| Elevators        | R   | 16599   | 18        | 19          | -             | <a href="https://openml.org/d/216">https://openml.org/d/216</a>                                                                                               |
| Yprop            | R   | 8885    | 251       | 0           | -             | <a href="https://openml.org/d/416">https://openml.org/d/416</a>                                                                                               |
| Topo             | R   | 8885    | 266       | 267         | -             | <a href="https://openml.org/d/422">https://openml.org/d/422</a>                                                                                               |
| SAT11            | R   | 4400    | 115       | 1           | -             | <a href="https://www.cs.ubc.ca/labs/algorithms/Projects/SATzilla/">https://www.cs.ubc.ca/labs/algorithms/Projects/SATzilla/</a>                               |
| Diamonds         | R   | 53940   | 6         | 3           | -             | <a href="https://openml.org/d/42225">https://openml.org/d/42225</a>                                                                                           |
| House_sales      | R   | 21613   | 20        | 1           | -             | <a href="https://openml.org/d/42731">https://openml.org/d/42731</a>                                                                                           |

## 8 ADDITIONAL RESULTS

### 8.1 TRAINING DATA

We use datasets from OpenTabs [Ye et al., 2024] for training, validation, and testing. We manually collect dataset descriptions and feature descriptions from UCI ML repository, Kaggle, OpenML, etc. Further, we curate metadata about the dataset by obtaining statistics about the dataset using python functions, for example, collecting potential classes for each target, data type of the feature values etc. Tables which have too few rows and columns, which have unclear and invalid data, are dropped. We identify task as classification or regression based on target value types, followed by min-max normalization of continuous values.

**ASPIRE:** We use the following specifications while implementing ASPIRE [Brahmavar et al., 2025]: Set Transformer-based architecture with model dimension 768, 8 attention heads, and 32 inducing points. The intra-instance Set2Set module and inter-instance aggregator each use 2 ISAB layers. Feature descriptions and categorical values are grounded via a frozen BERT-base-uncased encoder. Continuous values are encoded using 256 Fourier features. The prediction heads use a 10-component Mixture-of-Gaussians (MoG) for continuous targets and cosine-similarity with a learned temperature for categorical targets. For few-shot inference, ASPIRE conditions on  $k$  labeled support examples.

**CM2:** We use the following specifications: number of transformer layers 3, attention heads 8, batch size 256, learning rate for finetuning  $3e-4$ , patience is 5. For performing few-shot learning, Ye et al. [2024] prescribes  $k$ -shot learning via finetuning the model with  $k$  examples sampled from the train set. We observe that this causes high-variance in the performance, therefore we report the mean of all metrics for CM2 few-shot learning experiments.

**TP-BERTa:** TP-BERTa [Yan et al., 2024] is initialized from a RoBERTa backbone adapted for tabular prediction and is designed for standard supervised tabular prediction via fine-tuning on downstream datasets. Following the authors’ guidance, we train TP-BERTa per dataset with a larger training budget (up to 200 epochs) and early stopping with patience 50 epochs.

**TabSTAR:** We follow the TabSTAR specifications from [Arazi et al., 2026]. We use a pretrained encoder-only text model (e5-small-v2) to process categorical values and a two-layer MLP for the scalar numeric value that projects  $1 \rightarrow 2d \rightarrow d$  with ReLU and dropout.

### 8.2 EVALUATION DATA

The following table 3 details information about the downstream datasets we use as test datasets in our evaluation.

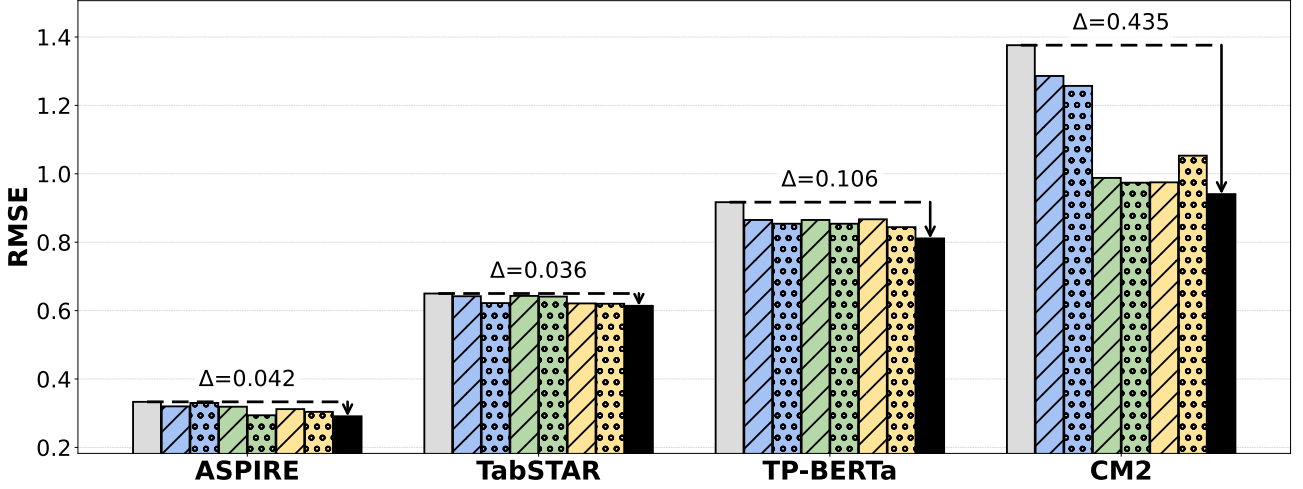


Figure 10: **Five-shot RMSE comparison** across data expansion strategies over various OCM backbones.

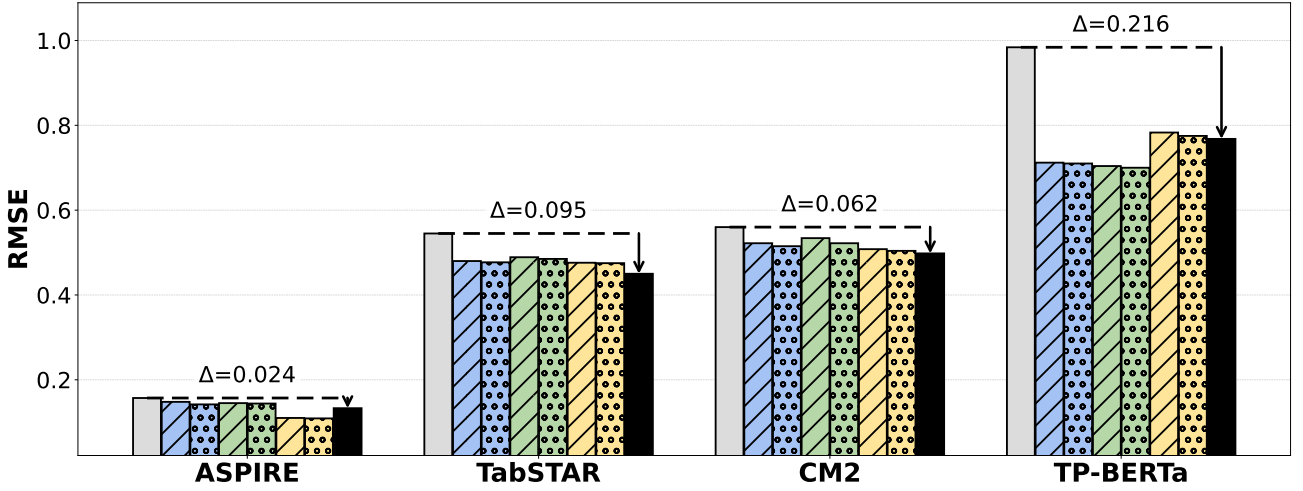


Figure 11: **Many-shot RMSE comparison** across data expansion strategies over various OCM backbones.

## 9 ADDITIONAL DETAILS

In this section we detail the configuration and prompts to LLMs used in different synthetic data generators like in our OUR TASK EXPANSION, CLLM, STRUCTSYNTH, TABPFN-BASED, and VANILLA LLM.

### 9.1 OUR TASK EXPANSION

**Models.** We used the following instruction-tuned LLM backbones for synthetic dataset specification generation: Llama-3.3-70B-Instruct, Meta-Llama-3.1-70B-Instruct, Meta-Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, QwQ-32B (and Preview), Qwen3-32B, Qwen3-30B-A3B-Thinking-2507, and gpt-oss-20b.

Each model independently generated a batch of synthetic datasets. The final corpus is the union of all datasets produced by these models. No model ensembling or cross-model mixing was performed. All models were used with the same decoding configuration. In this section, we mention additional results and details regarding data pre-processing and pretraining of the models.

**Walkthrough Example.** We trace the *rural diabetes screening* example (Figs. 2 and 3) through the TEXR pipeline. (1) *Task expansion*: the domain *Health & Sciences* produces the topic *rural diabetes screening*; the schema-generation

prompt then yields a context  $\tilde{S}$  with a dataset description (“Dataset for screening diabetes risk in rural populations...”) and features, e.g., *BMI* (continuous, [15, 50]), *age* (continuous, [18, 90]), and *diagnosis* (categorical, {positive, negative}). (2) *Weak instantiation*: the LLM proposes a DAG in topological order, e.g.,  $\text{age} \rightarrow \text{BMI} \rightarrow \text{diagnosis}$ , with conditional probability tables per node (quartile-conditioned Gaussians for BMI; a pmf over {positive, negative} conditioned on parent quartiles for diagnosis); topological sampling yields rows  $\tilde{X}$ . (3) *Filtration and cross refinement*: rows are filtered by the *z*-score criterion (Eq. 4), and values are revised by the cross-partition model that never trained on this dataset, e.g., adjusting implausible diagnosis values for low-BMI, young instances. The refined dataset joins the consolidated training corpus.

**Decoding configuration.** All models were accessed via `HuggingFace transformers` using chat-style prompting when supported. Generation used nucleus sampling with temperature 0.7,  $\text{top-}p = 0.95$ , and a maximum of 2048 new tokens (input truncated to 4096 tokens). We enforce JSON-only outputs and validate responses using a JSON parser; failed generations are retried up to three times.

**Prompt design.** The model is instructed to generate a fixed number of specific and realistic domains for a given category, returning the result strictly as a JSON array.

#### System Prompt

You MUST respond with ONLY valid JSON. No explanations, no extra text, no markdown.  
Start immediately with [ or {. End with ] or }. Nothing else.

#### Domain Generation Prompt

Generate exactly {count} specific business domains/industries for the category: {category}

Requirements:

- Return ONLY a JSON array of domain names
- Use specific, descriptive domain names (e.g., "telemedicine" not just "healthcare")
- Avoid generic terms, be specific and modern
- Include both traditional and emerging sub-domains
- Make each domain unique and realistic

Format: ["domain1", "domain2", "domain3"]

Generate {count} domains now:

#### Topic Generation Prompt

Generate exactly {count} dataset topics for: {domain\_1}, {domain\_2}, {domain\_3}

Format: ["topic1", "topic2", "topic3"] Examples: ["customer\_churn", "sales\_data", "inventory\_tracking"]

Generate {count} topics:

### Dataset Specification Prompt

For topic '{topic}', create dataset specification.

Return this JSON format:

```
{
  "description": "Dataset description",
  "features": [
    {"name": "age", "type": "continuous", "description": "Age of the subject in years", "range": [18, 80]},
    {"name": "category", "type": "discrete", "description": "Classification category", "categories": ["A", "B", "C"]}
  ]
}
```

Generate 8-12 realistic features for '{topic}'. Each feature MUST have a description explaining what it represents.

### BN Structure Prompt

Create edges with for Bayesian Network with features in a topological order: {feature\_names}

Return edge list as JSON:

```
[["parent1", "child1"], ["parent2", "child2"]]
```

### Discrete CPT Prompt

Generate probabilities for '{node}' categories: {categories}

Return JSON:

```
{"cat1": 0.0, "cat2": 0.0}
```

Make probabilities sum to 1.0:

### Continuous CPT Prompt

For feature '{node}' in range [{min}, {max}], parents: {parent\_config}.

Return probabilities for 4 quartiles as JSON array [p1, p2, p3, p4].

Probabilities must sum to 1.

## 9.2 TABPFN-BASED

Following the procedure in § 4 to generate data using TabPFN-based data expansion. We leverage the structural causal model (SCM) priors from TabPFN [Hollmann et al., 2023] to generate synthetic tabular datasets. Specifically, we sample from two SCM families—MLP-SCM and Tree-SCM—each parameterized with a random number of features (10–50), samples (500–2,000), and seed, producing feature matrices and continuous values. For classification tasks, we discretize the values using quantile binning to yield a mix type. Following is the prompt to add metadata and leverage a panel of LLMs-as-judge to filter out metadata. The final metadata is selected by max voting of the judge scores. We use Meta-Llama-3.1-70B-Instruct, Meta-Llama-3.1-8B-Instruct, Qwen3-32B, Mixtral-8x7B-Instruct-v0.1 models as judges.

## Prompt to generate Metadata

You are generating realistic metadata for a tabular dataset that should sound like it comes from a real-world domain.

**IMPORTANT:** Generate realistic, human-readable metadata that describes what each feature represents in real-world terms.

Requirements:

- Feature names should be MEANINGFUL and domain-appropriate (e.g., "patient\_age", "temperature", "account\_balance", "risk\_category", "payment\_status")
- Feature descriptions should be RICH and explain WHAT the feature measures in real-world terms:
  - \* For example continuous: "Patient age at time of hospital admission", "Monthly account balance in customer's primary checking account", "Heart rate measurement during initial assessment"
  - \* For example categorical:  
  
"Customer risk classification based on credit history and payment behavior",  
"Employment status indicating current work situation",  
"Product category for the transaction"
- For categorical features, explain what the categories represent and how they're used
- Dataset description should tell a compelling story: what domain, what purpose, what decisions it supports, time period, data source
- Do NOT include units - just describe what is being measured
- Do NOT mention standardization, normalization, or z-scores in descriptions
- focus on the real-world meaning
- Make descriptions informative and contextual (2-3 sentences for dataset, 1-2 sentences per feature)

Return STRICT JSON matching this schema (no markdown, no extra text):  
{schema}

EVIDENCE: TABLE  
{table}

EVIDENCE: TOP FEATURES FOR TARGET (mutual information)  
{top\_features\_for\_target}

EVIDENCE: FEATURE SUMMARIES (truncated)  
{features}

EVIDENCE: NEIGHBOR RELATIONSHIPS (correlation-based)  
{neighbors}

### Prompt to LLM-as-judge

```
You are an impartial judge evaluating metadata quality.

Return STRICT JSON matching this schema (no markdown, no extra text):
{
  "candidate_id": "string",
  "scores": {
    "evidence_consistency": "0-10",
    "specificity": "0-10",
    "helpfulness": "0-10",
    "conciseness": "0-10"
  },
  "total": "0-10",
  "flags": ["string"],
  "fix_suggestions": ["string"]
}

CANDIDATE METADATA (to evaluate)
{candidate_metadata}
Scoring guidelines:
- evidence_consistency: penalize any claim that
conflicts with numeric types/ranges/stats.
- specificity: reward descriptions that use evidence
(ranges, distributions, relationships) without overfitting.
- helpfulness: reward metadata that would help a model/user interpret features.
- conciseness: reward clarity without unnecessary verbosity.
```

## 9.3 CLLM

As prescribed by the authors, we use Mixtral-8x7B-Instruct-v0.1 as the LLM here to prompt and generate synthetic data with temperature  $T=0.9$ . For the curation step we using XGBoost model with  $n\_estimators=100$  trained on a sample of OpenTabs dataset and used to characterise each synthetic example by its confidence and aleatoric uncertainty.

### Prompt

```
You are a synthetic data generator.
Your goal is to produce as diverse samples as possible.

Leverage your knowledge of \textit{[dataset description]}
to generate 1000 realistic but diverse samples.

\texttt{\{format\_instructions\}}
The output should be a markdown code snippet
formatted in the following schema.
```json
{
  "feature\_1": string
  "feature\_2": string
  ...
  "target": string
}
```

## 9.4 STRUCTSYNTH

We retain the original two-stage StructSynth algorithm unchanged. To enable zero-shot operation, we introduce an upstream zero-shot seed generation phase that constructs synthetic dataset specifications, seed rows, and summary statistics, which are then supplied to StructSynth in zero-shot mode.

### 9.4.1 0-SHOT SEED GENERATION OVERHEAD

The upstream zero-shot generation phase follows the same task-expansion paradigm introduced earlier, extending it from specification generation to full dataset and summary-statistics construction.

**Domain and Topic Universe.** We first generate a universe of domains and dataset topics via LLM prompting. Each dataset is defined by: (i) domain, (ii) topic, (iii) a structured dataset specification containing 6–10 features with mixed types.

**Seed Dataset Construction.** For each specification, the LLM generates  $n = 20$  seed rows subject to strict schema constraints. All values are validated and normalized.

**Summary Statistics Generation.** We additionally prompt the LLM to produce realistic population-level summary statistics, including:

- categorical probability distributions,
- continuous mean/std/range parameters,
- pairwise association scores in  $[0, 1]$ .

If summary statistics are incomplete or invalid, deterministic fallbacks are applied using empirical estimates from the seed data.

### 9.4.2 STRUCTSYNTH IN ZERO-SHOT MODE.

The generated summary statistics are passed to StructSynth in zero-shot mode. In particular, statistical association scores used during dependency discovery are derived from the provided summary statistics rather than computed directly from data. In this configuration:

- schema inference uses provided summary statistics,
- association scores are replaced by summary-based values,
- no real dataset beyond the synthetic seed is required.

StructSynth then generates  $n_{\text{synth}} = 200$  structured samples per dataset using the original two-stage algorithm.

**Configuration.** All stages use `gpt-4o-mini` as the backbone model. Seed generation uses temperature 0.7. StructSynth uses temperature 0.9. Few-shot size  $k = 20$ . Row-batch size for structured generation is 1 unless otherwise specified. All outputs are JSON-validated with up to three retries per call.

## 9.5 VANILLA LLM

**Model Configuration.** We use `Qwen/Qwen3-4B-Instruct-2507` served via `vLLM` with temperature 0.7,  $\text{top-}p = 0.95$ , maximum 512 new tokens per call, and maximum context length 1024. Rows are generated in batches of 8. We generate datasets with 100 samples per dataset under a fixed wall-clock budget of 48 hours on a single 16GB GPU, ensuring computational parity with other methods. Under this constraint, the vanilla LLM baseline produced a total of 227 distinct datasets. The domain universe size is 150 and topic universe size is 1000.

**Generation Procedure.** The vanilla baseline follows the same domain, topic, and dataset specification prompt templates described above for OUR TASK EXPANSION and the 0-shot upstream stage of STRUCTSYNTH. However, no structural modeling (e.g., dependency graphs, CPTs, or summary-based association scores) is applied. Synthetic rows are generated directly from the LLM conditioned only on the dataset specification and feature constraints.